Physics Artificial Superintelligence Concepts

by

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Physice ASI Concepts

Introduction

The primary objective of this project is to create a conceptual framework of Artificial Superintelligence (ASI) based on one unexplained observation in physics, before Einstein's theories of relativity. The application chosen for this tutorial is a simulation of events that actually happened preceding 1900, pertaining to the mystery of the precession of Mercury's planetary orbit around the sun. Could a physics ASI module solve this problem in the same way as the pre-1900 physicists? In this application, instead of attempting to implement real ASI, the potential solutions to the problem are hard coded, and selected by an LLM model. The technical architecture of this application consists of a React frontend, a FastAPI backend, Langraph AI agent, Ollama LLM, and LanceDB vector database.

Document Structure

This report is organized into distinct sections:

- **Section 1: Background on Mercury Precession** – Explains the discrepancy between the observed precession of the planet Mercury and the prediction of Newtonian mechanics.
- **Section 2: Application Architecture** – Explains the framework and limited scope of this program.
- **Section 3: React Frontend** – Demonstrates the software's functionality and user interface through detailed screenshots.
- **Section 4: ASI Thought Experiments** – Examines some issues with physics ASI.
- **Section 5: Conclusions, Limitations, and Extensions** – Outlines application limitations and possible extensions to approach a real physics ASI module.
- **Appendix A: Oribital Mechanics Fundamentals** – Reviews the foundational physics principles and background relevant to the application.
- **Appendix B: Mercury Precession Issue** – Repeats Section 1 but with more math details.
- **Appendix C: ASI and Maxwell's Equations from a Pre-1855 Knowledge Base: Literature Review** – Explores literature to check if ASI has tried to generate Maxwell's equations.

- **Appendix D: General Relativity Derivation of Mercury Precession** – Discusses how to derive the precession of Mercury based on General Relativity.
- **Appendix E: Deduction, Induction, Abduction** – Explains the differences between these three core methods of logical reasoning.
- **Appendix F: Ollama Installation** – Contains instructions to install Ollama as a Windows service.
- **Appendix G - Retrieval-Augmented Generation (RAG) memory** – Explains how retrieval-augmented generation is applied to the AI agent in this application.

Target Audience

This document is for individuals seeking to understand ASI, including:

- **Scientists** who use AI assistants for routine tasks and wonder whether ASI could one day replace them.
- **Industry Skeptics** who doubt ASI could solve even a single historical problem in physics.
- **STEM Students** who find ASI fascinating but need a simplified tutorial to understand how difficult it is to implement.
- **Investors** who want to understand how ASI actually works and are looking for proofs of concept.

Note: Google Gemini, Claude, and Meta AI were used to assist in drafting portions of this document.

Section 1 – Background on Mercury Precession

As Mercury orbits the Sun, its elliptical path slowly rotates or wobbles over time, a movement known as precession. Classical Newtonian physics accounted for about 532 arcseconds per century of this shift due to the gravitational pull of other planets, but historical observations showed a total rate of roughly 575 arcseconds per century. The Mercury **precession anomaly** is the discrepancy where the orbital perihelion of the planet **Mercury** shifted by an unexplained (575-532) **43 arcseconds per century** under Newtonian mechanics.

1. The Newtonian Ideal: Closed Ellipses

In Newton's theory, gravity is a central force obeying an exact inverse-square law. For two isolated point masses, the force F and potential V are:

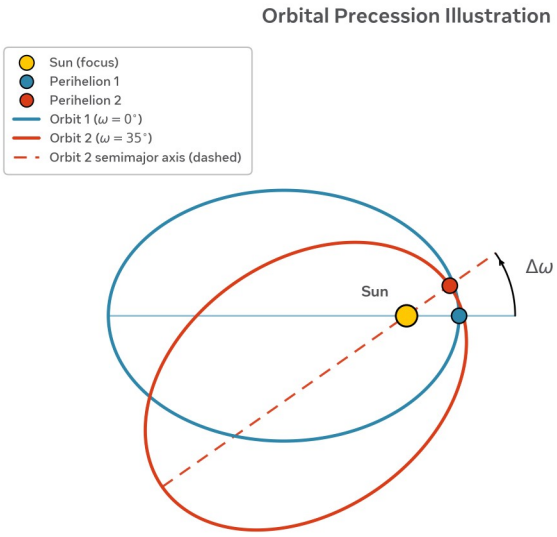
$$F = - Gm_1m_2 / r^2 \quad \text{and} \quad V(r) = - GM / r$$

For this potential, the Binet orbital equation (Appendix A) is exactly solvable. The solution is a fixed ellipse with semi-major axis a, eccentricity e, and a fixed argument of perihelion ω. The perihelion and aphelion always occur at the same angular positions. The orbit closes after 360°, which means there is no precession if there are no other disturbances. However, any deviation from $1/r^2$ must cause the ellipse to rotate — to precess.

2. What Is Perihelion Precession?

Each planet exerts a small perturbing pull on the other planets. The effect averages to a slow rotation of the whole ellipse in its own plane. After one orbit, Mercury does not return to the same perihelion direction; the perihelion advances by a small angle Δω per revolution. Over centuries this accumulates.

The diagram below illustrates geometrically what happens when the whole planetary orbit precesses by angle Δω, but note that observations are made on the perihelion location, which is the point of closest approach to the sun. The diagram exaggerates all dimensions for visualizing the effect, which is actually very very tiny.



3. The Observed Anomaly

The total observed precession of Mercury's perihelion was measured throughout the 19th century. Urbain Le Verrier in 1859 found that Newtonian perturbations could only explain 526.7″/cy of the 565″/cy then measured. The leftover 38″/cy (later refined by Simon Newcomb in 1882 to **43″/cy**) was too large to be observational error.

4. Why Was 43″ So Serious?

43″ per century is tiny — about 0.012° per century. Why did it matter? First, precision astronomy in the 19th century could measure angles to <1″. Second, Newtonian gravity had passed every other test to <1%. Third: if theory and observation disagree, either add unseen mass or fix the force law.

5. Why Try to Modify Newton's Gravity Before 1900?

Physicists before Einstein had a methodological toolkit:

1. Conservatism: Keep $F = Gm_1m_2 / r^2$ but add new sources. This is how Neptune was found.
2. Minimal modification: If no new mass is found, modify the analytic form of F(r) or V(r) slightly while preserving conservation laws.
3. Mechanistic spirit: Many physicists believed gravity must be mediated by an ether, like electromagnetism. It was natural to try similar terms for gravity.
4. Mathematical freedom: Nothing in Principia forbids $F \propto 1/r^{2+\delta}$ except simplicity. Testing δ is a legitimate experiment.

6. Pre-1900 Attempts to Save Newtonian Gravity

6.1. Intra-Mercurial Matter: Vulcan

Le Verrier's first proposal: a planet or ring inside Mercury's orbit, dubbed Vulcan.

Systematic searches during eclipses (1878, 1883, 1887) found nothing. By 1900 the hypothesis was largely abandoned.

6.2. Hall's Modified Power Law (1894)

Asaph Hall, discoverer of Mars' moons, proposed that gravity is not exactly inverse-square but:

$$F(r) = Gm_1m_2 / r^{2+\delta}$$

where δ is a tiny constant.

To obtain $\Delta\omega = 43''$ per century, we need:

$$\delta = \Delta\omega / 180° \approx 1.6 \times 10^{-7}$$

This reproduced Mercury but was ad hoc: it predicted precession for the lunar apogee and for Mars that was not observed, and violates energy conservation. Newcomb showed in 1895 that a single δ cannot fit all planets.

6.3. Velocity-Dependent Potentials

Inspired by Weber electrodynamics, Tisserand (1872), Lévy (1890), Scheibner and others tried potentials that depended not only on distance r, but also on velocity $\dot{r}$:

$$V(r, v) = - GM / r \times [1 + \alpha (\dot{r}^2/c^2) + \beta (r \ddot{r} /c^2) + ...]$$

Lévy found his theory violated Newton's third law.

Paul Gerber in 1898 revived the idea that gravity propagates at finite speed c_g:

Gerber derived:

$$\Delta\omega = 6\pi GM / [a (1 - e^2) c_g^2] \text{ per orbit}$$

Setting $c_g = c$ gives $\Delta\omega = 43''$/cy identically — same as Einstein later. However, historians (Roseveare 1982) have shown Gerber's derivation was flawed.

7. Why All Newtonian Modifications Failed

All pre-1900 patches faced three criteria: (i) reproduce Mercury's 43″, (ii) not spoil agreement for Venus, Earth, Mars (<10″/cy), and (iii) respect conservation laws. Hall's single δ failed (ii). Velocity-dependent laws failed (iii). Vulcan failed observationally. The core issue: Newtonian gravity assumes instantaneous action and Euclidean space. No algebraic fudge of $1/r^2$ can naturally produce the correct precession rate that was derived later by Einstein.

8. Epilogue: Einstein (1915)

Einstein's General Relativity replaces V = -GM/r with spacetime curvature (Appendix D).

$$\Delta\omega_{GR} = 6\pi\, GM_{Sun} / [\, a\,(1 - e^2)\, c^2\,] \quad \text{per orbit}$$

For Mercury this gives $\Delta\omega = 42.98''$ per century, exactly the anomaly.

Moreover, the same formula correctly predicts 8.62″/cy for Venus, 3.84″/cy for Earth, and 1.35″/cy for Mars — all confirmed today.

Thus, the 19th-century attempts to modify Newton's law were not naive; they were rational extensions within the Newtonian paradigm, motivated by spectacular success of perturbation theory (Appendix A) . Their failure hinted that the paradigm itself needed replacing .

References

- **Einstein, A.** (1915). Erklärung der Perihelbewegung des Merkur aus der allgemeinen Relativitätstheorie. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften zu Berlin*, 1915(2), 831–839.
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- **Hall, A.** (1894). A suggestion in the theory of Mercury. *The Astronomical Journal*, 14(14), 49. https://doi.org/10.5479/ADS/bib/1894AJ.....14...49H
- **Le Verrier, U.** (1859). Théorie du mouvement de Mercure. *Annales de l'Observatoire de Paris*, 5, 1–195.
- **Newcomb, S.** (1882). Discussion of observed transits of Mercury, 1677–1881. *Astronomical Papers Prepared for the Use of the American Ephemeris and Nautical Almanac*, 1(6), 363–487.
- **Roseveare, N. T.** (1982). *Mercury's perihelion: From Le Verrier to Einstein*. Oxford University Press.

Section 2 - Application Architecture

Instead of implementing a full physics ASI module, this application offers a basic framework to demonstrate the concept. The precession anomaly of the planet Mercury is used to illustrate what an ASI module might be required to do to match what physicists actually tried prior to 1900. A real ASI would implement a process of gravity model selection. Here, we just hard code 10 possible gravity models, and let an LLM perform the selection, without any ASI.

Candidate gravity model families:

inverse_cube_const: $F = -GM/r^2 - k/r^3$
inverse_cube_linear: $F = -GM/r^2 - \alpha \cdot GM/r^3$
inverse_cube_quadratic: $F = -GM/r^2 - \beta \cdot (GM)^2/r^3$
inverse_fourth_const: $F = -GM/r^2 - k/r^4$
inverse_fourth_linear: $F = -GM/r^2 - \alpha \cdot GM/r^4$
inverse_fourth_quadratic: $F = -GM/r^2 - \beta \cdot (GM)^2/r^4$
hall_law: $F = GM/r^{(2+\delta)}$
log_suppressed: $F = GM / [r^2 \ln(r/r_0)]$
yukawa: $F = -(GM/r^2)(1+\alpha r)e^{(-\alpha r)}$
gerber: $\Delta\varpi = 6\pi GM / [c^2 a(1-e^2)]$

For this scenario, each selection is evaluated against various factors and scored. The objective is to demonstrate the processes of ASI. Rather than 10 samples, a real ASI might try 100 or 1000 different variations, before finding one that satisfies all physical constraints. Even then, it would be limited by the math available at that time, pre-1900.

An AI agent is constructed to demonstrate the iterative logic flow that uses an LLM to select the next possible approach to solve the problem. An evaluation node uses stubs to mimic the basic steps a real ASI node would perform (as described below).

The Pre-1900 Gravity Model Selection agent implemented here is a LangGraph StateGraph — a directed graph in which each node is a Python function that reads from and writes back to a shared state dictionary (MDLState). LangGraph handles the traversal: after each node returns its updates, the framework merges them into the state and follows the outgoing edge or conditional routing function to decide which node runs next. No node calls another directly; all coordination is through the graph structure.

The six nodes divide cleanly into three categories: real retrieval and memory (retrieve_node, store_node), real evaluation and reflection (evaluate_node, reflect_node), and the LLM-driven proposal (propose_node). Two documented stub calls inside evaluate_node represent the parts

of a genuine automated-discovery pipeline that are not implemented here. Each is described in detail below.

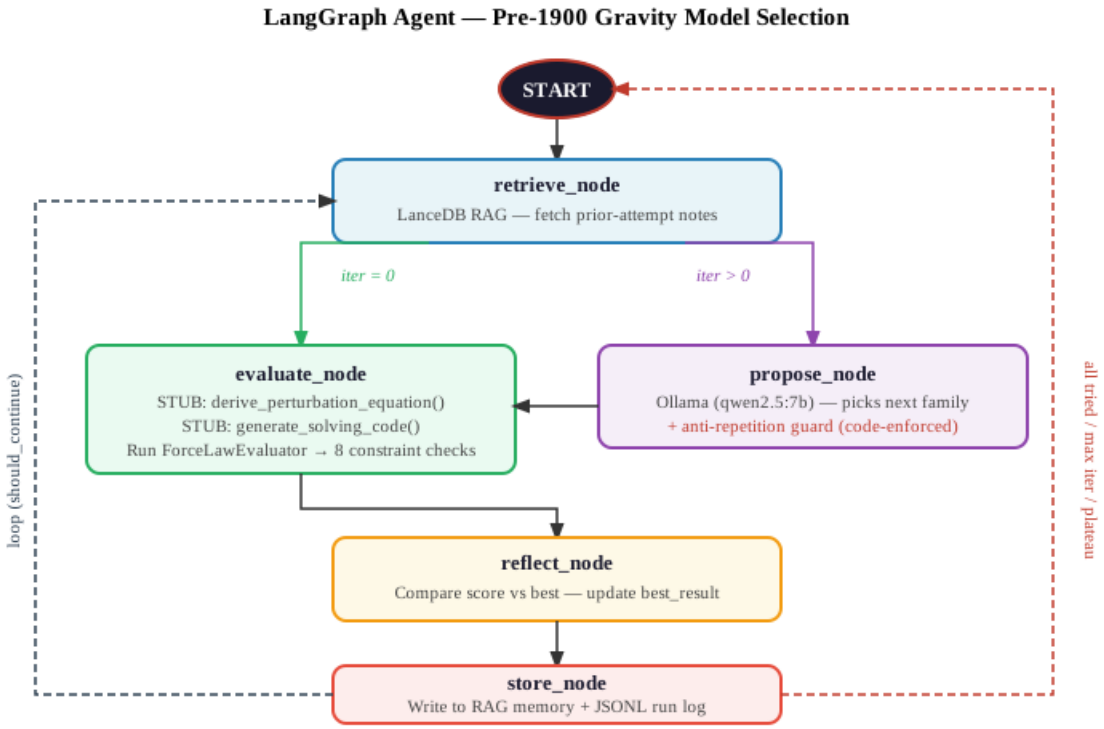


Figure 1. LangGraph state-graph for the Pre-1900 Gravity Model Selection agent. Dashed arrows show conditional routing; solid arrows are unconditional edges. The green path (iter = 0) skips propose_node so the user's chosen starting family is always evaluated first.

Node 1 — retrieve_node

retrieve_node is the entry point for every iteration. It queries the agent's RAG memory (a LanceDB vector store, embedded via Ollama's nomic-embed-text model) with the name of the current force-law family and returns the three most semantically similar past-attempt summaries (Appendix G). These notes are then available to the LLM in propose_node, giving the agent access to what worked and what did not in prior iterations without requiring the full history to fit in the prompt.

- Input state fields read: config.family
- Output state fields written: retrieved (list of text summaries)
- Real or stub: real — this is a genuine vector-similarity retrieval call

- On the very first iteration of a brand-new run, the LanceDB table does not yet exist; retrieve_node handles that gracefully and returns an empty list rather than raising an error.

Routing — route_after_retrieve

After retrieve_node, a conditional routing function (not a node itself) checks the current iteration counter. If this is iteration 0, control passes directly to evaluate_node, skipping propose_node entirely. This guarantees that whatever family the user selected in the UI is evaluated first, before the LLM ever makes a proposal. From iteration 1 onward, control passes to propose_node instead.

Node 2 — propose_node

propose_node calls a locally-running LLM (qwen2.5:7b via Ollama) and asks it to nominate the next force-law family to evaluate. The prompt includes: the current family and its scores, the best result found so far this run, a full list of already-tried families with their scores, a list of not-yet-tried families, and the retrieved notes from retrieve_node. The LLM responds with a JSON object naming the next family and a one-sentence rationale.

- Input state fields read: config.family, metrics, retrieved, best_result, history
- Output state fields written: config.family (the new proposal), proposal_raw
- Real or stub: real — this is a genuine LLM inference call

Anti-repetition guard (code-enforced, not prompt-only): after the LLM responds, propose_node checks whether the proposed family has already been tried this run. If it has, and untried families remain, the proposal is silently overridden with the lowest-complexity untried family, and the override is noted in the rationale field. This is enforced in Python, not in the prompt, because testing with qwen2.5:7b showed the model would repeat already-tried families despite explicit prompt instructions to the contrary.

Node 3 — evaluate_node ★ contains two stub calls

evaluate_node runs the physics evaluation for the proposed family. Its real work is calling ForceLawEvaluator.evaluate(), which fits the family's single free parameter to Mercury's observed 43″/century precession, then checks the same fitted parameter against eight held-out

constraints: precession predictions for Venus, Earth, Mars, and the Moon, plus Kepler's third law deviations for the same four bodies.

Before those real checks run, two documented stub functions are called at exactly the point a genuine automated-derivation pipeline would call them:

- **derive_perturbation_equation(family_name, force_law_formula) [STUB]**: *In a real ASI-for-physics system, this step would take the proposed force law (e.g. $F = -GM/r^2 - k/r^3$) and symbolically derive the perturbation equation that describes how that extra term causes orbital precession. This requires choosing the right mathematical machinery: substituting into the Binet equation, identifying the secular (resonant) driving term, and reducing the equation to a solvable form. It is an abductive step — proposing which formalism applies to a novel input — which current LLMs cannot do reliably for genuinely novel force laws (see Zahavy, "LLMs Can't Jump"). In this project, all ten perturbation equations were derived by hand and cross-checked independently before being trusted; this stub marks where that derivation would be automated. Returns None.*

- **generate_solving_code(derived_equation) [STUB]**: *In a real pipeline, once the perturbation equation exists symbolically, this step would generate executable Python code to evaluate it numerically for any given orbital parameters. This is closer to something current LLMs can do (translating a known closed-form expression into code is more deduction than abduction), but because derive_perturbation_equation above returns None, there is nothing real to translate here either. A genuine implementation would call an LLM code-generation endpoint, then execute the output in a sandboxed environment before trusting it — never directly. In this project, the hand-written predict_precession and kepler_deviation methods in models/families.py serve the role this generated code would play. Returns None.*

Because both stubs return None, evaluate_node proceeds with the pre-computed formulas already built into each ForceLawFamily subclass — exactly as if a real pipeline had produced them. The stubs exist so the shape of the full derive-then-code pipeline is visible in the running code path, not merely claimed in a comment.

- Input state fields read: config.family
- Output state fields written: metrics, formula, note, fit_params, precession_checks, kepler_checks

Node 4 — reflect_node

reflect_node compares the current iteration's overall_score against the best score found so far this run (tracked in best_result, not inferred from the history list). If the new score is better, best_result is updated and the plateau counter is reset to zero. If not, the plateau counter increments. The plateau counter is used by store_node to decide whether to stop early once all families have been exhausted.

reflect_node also calls llm.summarize_reflection(), which produces a one-line human-readable summary of whether this iteration improved on the best result or not. This summary is stored in the run log and displayed in the frontend's iteration log panel. Unlike propose_node, this is a deterministic string-formatting call, not an LLM inference call.

- Input state fields read: metrics, best_result, config.family, formula, note, precession_checks, kepler_checks
- Output state fields written: reflection, plateau_count, best_result (when improved)
- Real or stub: real — no stubs here

Node 5 — store_node

store_node writes the completed iteration to two persistent stores and then decides whether the loop should continue. The two stores serve different purposes: MDLMemory (LanceDB) stores a free-text embedding of the attempt for semantic retrieval in future iterations; the JSONL run log (logs/runs/run_<timestamp>.jsonl) stores a structured, replayable record of every iteration in the run, independent of both the frontend's in-memory state and the RAG memory.

The continuation decision follows three rules, checked in order: (1) if the iteration counter has reached max_iterations, stop; (2) if every family in the fixed menu of ten has been tried at least once, stop cleanly — there is nothing new to propose; (3) if every family has been tried and the plateau counter has reached 3 consecutive non-improvements, stop. The plateau check is deliberately disabled while untried families remain, because the agent should explore all candidates even if recent proposals happened to be worse than the current best.

- Input state fields read: history, config, metrics, reflection, iteration, max_iterations, plateau_count
- Output state fields written: history (appended), iteration (incremented), should_continue
- Real or stub: real — no stubs here

Routing — route_after_store

After store_node, a second conditional routing function reads should_continue. If true, control returns to retrieve_node for the next iteration. If false, the graph terminates at END and the backend returns the final state (including best_result and the complete history) to the frontend.

Why Ollama? What is LangGraph doing?

Ollama is a local LLM inference server. This project uses it for two purposes: running qwen2.5:7b to make family proposals in propose_node, and running nomic-embed-text to compute vector embeddings for the RAG memory in retrieve_node and store_node. Both run entirely on the local machine; no data leaves the user's computer. Ollama is required because the agent needs a live LLM call at proposal time — it cannot be pre-computed.

LangGraph is the orchestration framework. It is not doing anything the Python function calls could not do directly, but it provides three practical advantages: (1) the graph structure is explicit and inspectable, making the flow visible rather than buried in a while loop; (2) state updates are merged cleanly without every node needing to copy the whole state dict; (3) conditional routing is declared once at the graph level rather than scattered through the function bodies.

The agent is not simulating artificial superintelligence — it is a constrained selection loop that chooses among ten pre-derived candidates. The stubs in evaluate_node mark exactly where the boundary is between what this demo implements and what a genuine automated-discovery system would need to add.

References:

https://medium.com/@martin.hodges/defining-the-langgraph-state-47c5ef97a95c

https://devops.gheware.com/blog/posts/langgraph-stategraph-beginners-guide-2026.html

https://realpython.com/langgraph-python/

Section 3 - React Frontend

Overview of application functionality with screenshot annotations.

Running the Application

When you download a PowerShell script (`.ps1`) from the internet, Windows automatically flags it with an **"Internet Zone" identifier**. If your system's execution policy is set to `RemoteSigned` (the Windows default), PowerShell will **strictly block unsigned scripts** carrying this internet flag from running.

Open Powershell (non-admin), go to physics-asi-concepts\scripts directory, and to unblock the script named ‘unlock.ps1’, enter:

```
Unblock-File .\unlock.ps1
```

Then, to unblock the other powershell scripts, enter:

```
.\unlock.ps1
```

The Windows PowerShell script ‘install.ps1’ installs all required application components. However, because Ollama requires a download of approximately 5 GB, we recommend following the separate installation instructions for Ollama provided in Appendix F.

To setup and run the application see install.ps1, run.ps1, and stop.ps1.

Begin by entering:

```
.\ install.ps1
```

After that script completes, enter:

```
.\run.ps1
```

The script ‘run.ps1’ will automatically open a browser window with a user interface to the application.

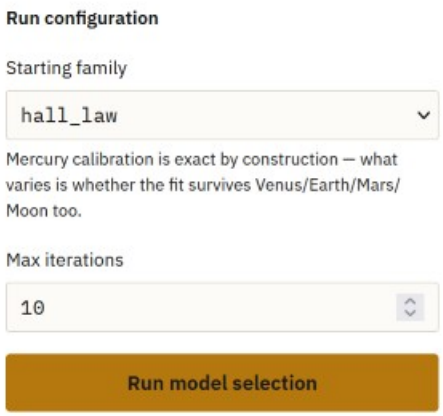
After the application has completed, to close the background processing, you must enter:

```
.\stop.ps1
```

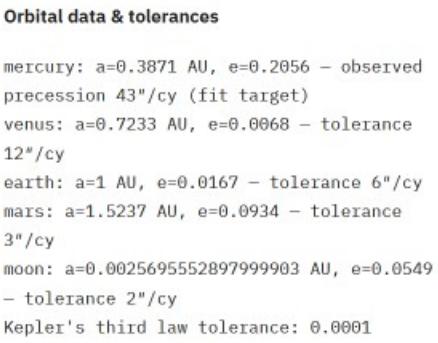
Overview

The screenshots begin with "Run configuration", which allows the user to select a starting family, and the max iterations to find a suitable perturbation to Newton's inverse square law of gravity. On each iteration, the agent proposes a candidate force-law family — either the one selected in "Starting family" on the first pass, or a new family chosen based on the previous iteration's outcome — and checks it against the orbital precession and Kepler's-third-law tolerances shown in the reference panels. If the candidate passes every check, it becomes the new "best result"; if it fails or simply scores worse than the current best, it's logged as the "latest attempt" but the best result is left untouched. Either way, the agent then reflects on the outcome — noting which constraints passed or failed and by how much — and uses that reflection to decide whether to retry the same family with adjusted parameters, move to a different candidate family, or stop early if a family already satisfies all tolerances with margin to spare. This propose–evaluate–reflect cycle repeats until either a satisfactory fit is found or the run reaches the configured max iterations, at which point the best result found across all iterations is reported.

The control panel where the user picks a starting force-law family and the maximum number of iterations, then launches a run.



Reference panel listing the real orbital elements and precession tolerances each candidate family is checked against.



Reference panel listing the pre-1900 physics and mathematics available to the agent — nothing from after 1900 is used anywhere in the pipeline.

Known physics & math (pre-1900 input)

```
1687: Newton's Laws of Motion (F = ma)
1687: Newton's Law of Universal
Gravitation (F = GMm/r²)
1609: Kepler's First Law (r(θ) = p / (1 +
e·cosθ))
1609: Kepler's Second Law (dA/dt =
constant)
1619: Kepler's Third Law (T² ∝ a³)
1687: Differential and Integral Calculus
1750: Ordinary Differential Equations
1799: Perturbation Theory
1811: Binet's Equation
1687: Newton's Apsidal Precession Theorem
```

Reference panel listing all ten candidate force-law families the agent can choose among, with their formulas and complexity scores.

Candidate families

```
inverse_cube_const: F = -GM/r² - k/r³
(struct=3, params=1)
inverse_cube_linear: F = -GM/r² - α·GM/r³
(struct=4, params=1)
inverse_cube_quadratic: F = -GM/r² -
β·(GM)²/r³ (struct=5, params=1)
inverse_fourth_const: F = -GM/r² - k/r⁴
(struct=3, params=1)
inverse_fourth_linear: F = -GM/r³ - α·GM/
r⁴ (struct=4, params=1)
inverse_fourth_quadratic: F = -GM/r² -
β·(GM)²/r⁴ (struct=5, params=1)
hall_law: F = GM/r^(2+δ) (struct=3,
params=1)
log_suppressed: F = GM / [r² ln(r/r₀)]
(struct=6, params=1)
yukawa: F = -(GM/r²)(1+αr)e^(-αr) (struct=6,
params=1)
gerber: Δϖ = 6πGM / [c²a(1-e²)] (struct=8,
params=0) — numerically fine,
theoretically ad-hoc
```

The best result found so far in the run (separate from the latest attempt): inverse_cube_quadratic passes every precession and Kepler's-third-law check.

Best result — inverse_cube_quadratic (iteration 3)

$$F = -GM/r^2 - \beta\cdot(GM)^2/r^3$$

PRECESSION (ARCSEC/CENTURY)

Venus	8.629 "/cy	tol 12.00	PASS
Earth	3.840 "/cy	tol 6.000	PASS
Mars	1.352 "/cy	tol 3.000	PASS
Moon	0.060 "/cy	tol 2.000	PASS

KEPLER'S THIRD LAW (FRACTIONAL DEVIATION)

Venus	8.19e-8	tol 1.00e-4	PASS
Earth	5.93e-8	tol 1.00e-4	PASS
Mars	3.89e-8	tol 1.00e-4	PASS
Moon	6.93e-11	tol 1.00e-4	PASS

The most recent attempt (gerber), shown in its own panel so a regression is still visible even while the best result stays on screen.

Latest attempt — gerber

$$\Delta\varpi = 6\pi GM / [c^2 a(1-e^2)]$$

numerically fine, theoretically ad-hoc

PRECESSION (ARCSEC/CENTURY)

Venus	8.625 "/cy	tol 12.00	PASS
Earth	3.839 "/cy	tol 6.000	PASS
Mars	1.351 "/cy	tol 3.000	PASS
Moon	0.060 "/cy	tol 2.000	PASS

KEPLER'S THIRD LAW (FRACTIONAL DEVIATION)

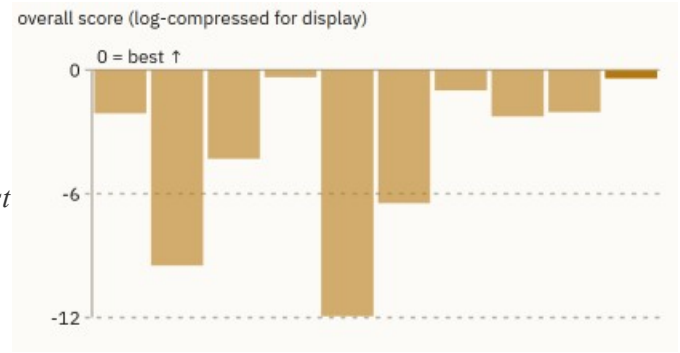
Venus	4.09e-8	tol 1.00e-4	PASS
Earth	2.96e-8	tol 1.00e-4	PASS
Mars	1.96e-8	tol 1.00e-4	PASS
Moon	3.47e-11	tol 1.00e-4	PASS

Accuracy penalty and complexity score plotted per iteration — the agent tries a worse family, then returns to the best one it already found.

Metrics over iterations



Overall score (log-compressed for display) for every family tried so far; 0 marks the unreachable ideal, and the latest bar is shown at full opacity.



The iteration log, recording which family was tried each round and whether it improved on the best result found so far.

Iteration log

0	family=hall_law -> new best (best so far -inf -> this round -70.439, accuracy_penalty=70.1890)	hall_law
1	family=inverse_cube_const -> no improvement over best (best so far -70.439 -> this round -1689045302.271, accuracy_penalty=1689045302.0209)	inverse_cube_const
2	family=inverse_cube_linear -> no improvement over best (best so far -70.439 -> this round -10017.210, accuracy_penalty=10016.9105)	inverse_cube_linear
3	family=inverse_cube_quadratic -> new best (best so far -70.439 -> this round -0.350, accuracy_penalty=0.0000)	inverse_cube_quadratic
4	family=inverse_fourth_const -> no improvement over best (best so far -0.350	inverse_fourth_const

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Section 4: ASI Thought Experiments

While this tutorial demonstrates a conceptual framework for implementing Artificial Superintelligence (ASI) logic for one particular physics problem, the question remains – could it be extended to more problems? Let's explore how ASI could have been applied across different time periods in physics. Without any actual ASI model, we can analyze its processes through a series of thought experiments. For each case, we ask: what capabilities would an ASI need to re-derive the result, and what specific reasoning tool is missing that prevents a purely data-driven system from succeeding?

Thought Experiment 1: From Kepler to Newton (1687)

Background: Kepler's third law gives $T^2 \propto a^3$, approximated for circular orbits as $T^2 \propto r^3$. Huygens (1659) derived centripetal acceleration $a_c = v^2/r$.

Derivation: In Principia (1687), Newton synthesized three results: Kepler III, Huygens' $a_c = v^2/r$, and his own $F = ma$. With $v = 2\pi r/T$, $a_c = 4\pi^2 r/T^2$. Substituting $T^2 \propto r^3$ yields $a_c \propto 1/r^2$, hence $F \propto m/r^2$. Then he used geometric analysis to determine the elliptical orbits of the planets.

What ASI Would Need to Succeed:
1. Symbolic Algebra Engine: manipulation of proportionalities and substitution.
2. Kinematic Modeling: ability to model circular motion as $v = 2\pi r/T$.
3. Inductive Generalization Tool: this is the missing piece. The data only show $T^2 \propto r^3$ for planets. The leap to $F \propto Mm/r^2$ for ALL masses — terrestrial and celestial — is an abductive universalization. No observation in 1687 showed Earth attracting apples with the same law as Sun attracting planets.

Missing Tool – Ontological Commitment and Universalization: An ASI that only fits planetary data would produce $F \propto C/r^2$ where C is a planet-specific constant. To obtain universal gravitation, it needs a tool for concept formation: hypothesizing that C factorizes into a product of masses ($C = G \cdot M \cdot m$), that mass is the source of the force, and that the same law applies universally. This requires a principle of universality and a mechanism to invent a new physical concept (universal mass attraction) not present in the input data.

Thought Experiment 2: Maxwell's Equations (1865)

Modern retrospective form (Heaviside-Gibbs 1884):
1. $\nabla \cdot E = \rho/\varepsilon_0$ – Gauss's law for electricity
2. $\nabla \cdot B = 0$ – Gauss's law for magnetism (no monopoles)
3. $\nabla \times E = -\partial B/\partial t$ – Faraday-Maxwell law (1831)
4. $\nabla \times B = \mu_0 J + \mu_0 \varepsilon_0 \partial E/\partial t$ – Ampère-Maxwell law (1826 + displacement current 1861-62)

Historical Pathway: The first three laws generalized experiments by Coulomb, Ampère, Biot-Savart, and Faraday. The critical data point was Weber and Kohlrausch (1856-57): the ratio of electrostatic to electromagnetic units ~3.1×10^8 m/s, close to Fizeau's 1849 speed of light. Maxwell showed his equations predict waves at $c = 1/\sqrt{(\mu_0\varepsilon_0)}$.

What ASI Would Need to Succeed:
1. Field Representation Engine: multivariable calculus for divergence and curl, even though Maxwell used components and quaternions; modern ASI would need vector calculus.
2. Multi-source Data Integration: combining electrostatics, magnetostatics, and induction experiments into differential laws.
3. Theoretical Consistency Repair Tool – this is the missing piece. Ampère's law $\nabla\times B = \mu_0 J$ violates charge conservation, because $\nabla\cdot(\nabla\times B) = 0$ always, but $\nabla\cdot J = -\partial\rho/\partial t \neq 0$ for a charging capacitor. No experiment in 1860 directly showed $\partial E/\partial t$ creating B. Maxwell invented the displacement current $\mu_0\varepsilon_0\,\partial E/\partial t$ to restore consistency.

Missing Tool – Abduction for Consistency: An ASI that only interpolates data would never add the term. It needs a meta-logical drive to enforce mathematical consistency ($\nabla\cdot J + \partial\rho/\partial t = 0$) and symmetry, and to hypothesize an unobserved physical mechanism to fix the inconsistency. Additionally, it needs a Unification Tool: recognizing numerical coincidence ($1/\sqrt{(\mu_0\varepsilon_0)} \approx c$) as physical identity – that light IS the electromagnetic wave, requiring an ontological merge of optics and electromagnetism.

Also, see Appendix C.

Thought Experiment 3: Mercury's Anomalous Precession (1859–1900)

Background: Le Verrier (1859) reported Mercury's perihelion advances ~43 arcseconds/century beyond Newtonian perturbations from other planets, confirmed by Newcomb (1882 onward).

Pre-Einstein Attempts: Many ad hoc corrections were proposed of the form $V(r) = -GMm/r - \alpha/r^2$ or $V(r) = -GMm/r - \beta/r^3$. Each produces extra precession and can be tuned to 43″.

What ASI Would Need to Succeed:
1. High-Precision Orbital Mechanics Engine: n-body integration and perturbation theory to compute expected Newtonian precession.
2. Symbolic Regression Tool: search over potential functions to fit the 43″ residual. This tool exists – Hall (1894) found similar laws by hand.
3. Theory Evaluation Tool – this is the missing piece for true discovery.

Missing Tool – Distinguishing Phenomenological Fit from Physical Theory: An ASI with symbolic regression would easily discover $V = -GMm/r - \beta/r^3$ that fits Mercury. That is not a

discovery of new physics; it is curve-fitting. What is missing is a criterion for physicality: a tool to reject ad hoc, fine-tuned additions that explain only one anomaly and fail elsewhere (e.g., for other planets or for energy conservation). Newton's law was preferred because it was simple and universal. The anomalous term is not. To move beyond fitting, ASI needs a model selection principle grounded in explanatory power, simplicity, and universality – not just residual error minimization.

Thought Experiment 4: General Relativity (1910–1915)

Background: After confirming Mercury's anomaly, Einstein (1907) introduced the Equivalence Principle: inertial mass = gravitational mass (tested by Eötvös to high precision). Free fall is locally indistinguishable from inertial motion. Einstein could not proceed until Grossmann introduced him to Riemann's curved manifolds (1854) and Ricci-Levi-Civita's absolute differential calculus (1900). Entwurf theory (1913), final field equations November 1915.

What ASI Would Need to Succeed:
1. Tensor Calculus Engine: Riemannian geometry, Ricci calculus, manipulation of $G\mu\nu$, $T\mu\nu$.
2. Principle Enforcement Tool: impose general covariance and equivalence principle as constraints.
3. Variational Selection Tool: choose among infinitely many covariant actions.

Missing Tools – Three distinct gaps:

a) Physical Reification Tool (Math → Physics): The mathematics of curved manifolds existed before physics. The leap is asserting that spacetime curvature IS gravity – that free particles follow geodesics, not because a force acts, but because spacetime itself is curved. An ASI can manipulate $R\mu\nu$ but needs a tool for ontological reification – promoting a mathematical structure to a physical entity.

b) Abductive Action Selection Tool: Even with covariance, infinite Lagrangians are possible: R, R^2, $R\mu\nu R\mu\nu$, etc. Einstein selected the simplest – the Einstein-Hilbert action $\int R\sqrt{-g}\, d^4x$ – that yields second-order equations and reduces to Newtonian gravity in the weak-field limit. This requires a meta-principle combining simplicity, stability, and correspondence with prior theory. Pure covariance does not select it.

c) Equivalence-to-Geometry Bridge: The step from "inertial mass = gravitational mass" (an empirical equality) to "gravity is inertial motion in curved spacetime" is an abductive leap. No deduction forces it. ASI would need a conceptual abduction engine that proposes geometric explanation for a numerical equality, rather than treating it as coincidence.

Summary: What Prevents a Pure Data-Driven ASI?

Experiment	Can Data + Math Engine Do It?	Missing Reasoning Tool
1. Newton	Yes – derives $1/r^2$	Universalization & Concept Formation: invent mass as source and assert universality
2. Maxwell	Partial – first 3 laws	Consistency Repair & Unification: invent displacement current to fix conservation, identify light with EM wave
3. Mercury anomaly	Yes – finds $-\beta/r^3$ fit	Theory Evaluation: reject ad hoc fits via simplicity/universality criteria
4. GR	Math possible, physics not	Reification + Action Selection + Equivalence-to-Geometry abduction: turn math into physical ontology and choose simplest covariant action

Patterns

What is the pattern in these thought experiments?

1. Physics breakthroughs require simultaneous availability of specific mathematics and precise physical definitions. Newton required geometry + Kepler's empirical laws. If vector calculus had been invented earlier, Maxwell would have used that + the experimental laws of Coulomb , Ampère, Faraday, and Gauss. The anomaly issue required calculus and perturbation theory. Einstein required tensor calculus + equivalence principle.

2. Physics relies on operational definitions using both words and formulas. However, modern physics textbooks contain hundreds of such operationally defined concepts. If ASI generates abstract symbols α, β, ω, G without mapping them to a shared human ontology, humans cannot learn from it.

3. Discovery vs. naming: There is a difference between discovering a law of the form $da/dz = 0$ and recognizing it as "the gravitational acceleration near Earth's surface, g, is constant". The first is symbolic regression; the second is conceptual grounding.

Are we proposing ASI models that may actually learn new physical principles, but humans cannot interpret the results? Yes – this is the interpretability gap. Further research on symbol grounding and LLM-augmentation is required.

Stepwise Approach: Historically Grounded Curriculum

It seems crucial for an ASI Physics application to maintain three separate, versioned knowledge bases: (a) an up-to-date mathematics base (symbolic libraries + differential algebra, e.g. Rosenfeld-Gröbner simplification), (b) a physics concepts ontology (knowledge graphs mapping symbols to operational definitions and natural language names), and (c) human observations/experimental data with uncertainty modeling.

A stepwise, historically controlled approach may work as a proof of concept. Consider the sequence outlined above, but with explicit knowledge cutoffs based on dates: 1687, 1865, 1900, 1910. At each date, let a candidate physics ASI model attempt to solve the problem. After each cutoff date, begin updating the knowledge bases, until the next problem arises. Then repeat this process, step by step, to keep improving physical theories.

This curriculum tests both symbolic discovery (does the equation appear?) and conceptual grounding (can humans understand and name it?). If an ASI model could solve these four candidates, then perhaps we could agree that physics ASI has a future. What if ASI cannot solve the problem? If physicists must prompt the ASI module, then its not really ASI.

The thought experiments considered specific time periods in the history of physics. In practice the stepwise approach to ASI would be applied nearly continuously or soon after any new data or new issue arises from physics experiments. In this scenario ASI does not replace physicists, it speeds up their discoveries, step by step.

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Section 5. Conclusions, Limitations, and Extensions

5.1 Conclusions

This project demonstrates, in working code, the shape of an automated scientific-discovery pipeline applied to a real historical episode: selecting among competing modifications to Newtonian gravity using only physics and mathematics available before 1900. The LangGraph agent correctly reproduces the historical outcome — inverse_cube_quadratic, which reduces exactly to the general-relativistic weak-field precession formula, wins on a combination of accuracy and Minimum Description Length (MDL) complexity, with Gerber's velocity-dependent theory identified as the one genuine observational degeneracy in the candidate set. The anti-repetition guard, the plateau-stop condition, and the honest separation between real code and documented stubs (Section 4) all work as intended and were each verified against real run output, not just designed on paper.

At the same time, this is a deliberately small, constrained system, and it should not be read as a demonstration of artificial superintelligence (ASI) applied to physics. The sections below enumerate specifically where the simplification lives, and what a genuine ASI-for-physics module would need to add to close each gap. Several of the architectural ideas below are drawn directly from recent automated-discovery systems — AI-Newton, AI Feynman, PySR / SymbolicRegression.jl, SR-Scientist, and SymLang — rather than invented for this document; citations are given throughout.

5.2 Limitations of the Current Implementation

- **Fixed candidate menu, not open-ended discovery**: propose_family selects among ten pre-derived force-law families (models/families.py). It is a constrained selection problem and can never propose an eleventh, qualitatively new family. Nothing in the current system searches the space of possible force laws; it only searches the space of ten pre-written answers.
- **The two stubs are true no-ops**: derive_perturbation_equation and generate_solving_code (pipeline/derivation.py) both return None. All ten perturbation equations in models/families.py were derived and cross-checked by hand; no part of this project actually performs symbolic derivation or code generation at run time.
- **No real symbolic regression engine**: There is no component that fits an open-ended expression tree to data. The ten candidate formulas are hard-coded Python methods, not the output of a search algorithm.

- **No genuine physics-experiment knowledge base**: data/orbital_data.py supplies a small, fixed table of real but static historical values for five solar-system bodies. There is no database of raw, noisy, multi-experiment data across varied physical systems — they are assumed as given Newtonian primitives in knowledge_base/.
- **No uncertainty quantification over competing structures**: Gerber vs. inverse_cube_quadratic is a genuine case of two structurally different theories making numerically indistinguishable predictions (see README). The current system resolves this with a single deterministic complexity tie-break rather than reporting it as a structural degeneracy with calibrated confidence.
- **Shallow, unstructured memory**: The LanceDB RAG store (agent/memory.py) holds free-text summaries for similarity retrieval, not a structured, queryable graph of concepts, laws, and their dependencies.
- **Single LLM call per iteration, no self-verification loop**: propose_node makes one Ollama call and evaluate_node runs one deterministic evaluation; there is no long-horizon, tool-using agent that inspects its own intermediate results, revises a hypothesis, and resubmits it within the same iteration.

5.3 Extensions: Toward a Genuine ASI-for-Physics Module

Closing the gaps above is not one change but several semi-independent modules, each replacing a specific piece of the current pipeline. They are presented in the order a physicist would actually use them: data in, structure out, derivation, verification, and finally a reasoning module. Before reading this section, it is recommended to review the distinction between deduction, induction, and abduction (see Appendix E).

(a) Physics Experiment Knowledge Base

Replaces: the fixed table in data/orbital_data.py

Technical name: Mathlib + PhysLib + SciLean as ground truth; knowledge graph (59+ nodes, 115+ typed edges) with evidence labels and Lean target audit.

Mathlib is a community-built library that as of 2025 contains over 210,000 proven theorems. PhysLib is its physics cousin — about 180,000 lines of physics definitions and proofs, from classical mechanics to quantum field theory. Unlike Wikipedia, nothing gets in here unless a computer has checked the proof down to the last logical step. On top of that, there is a knowledge graph — imagine a giant mind-map where each bubble is a concept like 'entropy' or 'Schrödinger equation' and each arrow says HOW they are connected ('derived from',

'approximates', 'contradicts'), with a label saying whether the link is formally proven, from literature, or just a guess. This gives provenance: you can click any fact and see where it came from, who proved it, and when.

(b) ASI Symbolic Equation Module

Replaces: propose_family's fixed menu, and the empty generate_solving_code stub.

The core engine should be a genuine symbolic-regression search rather than selection among pre-written candidates — an evolutionary multi-population backend such as PySR / SymbolicRegression.jl (Cranmer, 2023). On top of that engine, SymLang's symmetry-constrained grammar (Baig & Gillani, 2026) should encode parity, rotational, and Galilean/Lorentz invariance as hard production rules, pruning physically inconsistent candidate force laws before fitting rather than discarding them after. Finally, SymLang's MDL-regularized Bayesian model selection with bootstrap stability analysis should replace the current single-winner tie-break, so that a genuine degeneracy — like Gerber vs. inverse-cube — is reported honestly as structural uncertainty with a confidence measure, instead of a single, possibly overconfident answer.

(c) ASI Math Module

Replaces: the derive_perturbation_equation stub.

This module needs an actual computer-algebra backend — such as the Rosenfeld–Gröbner algorithm from differential algebra — capable of substituting a candidate force law into the Binet equation, carrying out the perturbative expansion, extracting the secular (resonant) driving term, and reducing the result to a minimal symbolic form. This is the literal mechanical work the current stub skips. It is worth being precise about what kind of capability this is: deriving the consequences of an already-specified force law is a deduction task, not an abduction task. That is exactly why the counterpoint raised earlier in this document matters here — Zheng-Xin's (2026) argument, illustrated by Feynman's largely deductive reconstruction of General Relativity (Di Mauro, Esposito & Naddeo, 2021), is specifically that this kind of structured, consequence-deriving work is tractable automation.

(d) Agentic Verification Loop

Replaces: the single deterministic call inside evaluate_node.

Following SR-Scientist (Xia, Sun & Liu, 2026), the agent should be given actual code-interpreter tools — a data-analyzer tool and an equation-evaluator tool that fits constants with

BFGS(numerical method to find the minimum of an objective function) and reports fit quality — inside a genuine long-horizon ReAct (Reasoning and Acting) loop: propose, fit, inspect residuals, revise, resubmit, all within a single iteration rather than one LLM call per iteration. An experience buffer of the top-K equations found so far, fetched at the start of each iteration, should supplement (not replace) the existing free-text RAG memory.

(e) Concept-Discovery Layer

Add: plausible-reasoning layer.

The knowledge_base currently hands the agent Newton's laws, Kepler's laws, and the relevant mathematics as given. There is a discrepancy between the observations and the theory. This is where physics **abduction** happens. Given incomplete data, generate the most plausible law:

A. Lagrangian Hypothesis Generation: LLM + symbolic regression (PySR, SINDy) proposes candidate Lagrangians. For example, observing pendulum with damping, propose $L = 0.5\ m\ l^2\ \dot{\theta}^2 + m\ g\ l\ \cos\theta + \lambda\dot{\theta}$.

B. Dimensional & Symmetry Check (fast filter): Every proposal passes through: dimensional analysis [Buckingham Pi], Lorentz / Galilean invariance, gauge invariance, parity/time reversal if required. 90% of LLM proposals die here — cheap and powerful.

C. Plausibility Scoring via Action Principle: Score = -log P(data|law) + α * complexity(law) + β * symmetry_break_penalty. Where -log P(data|law) is simulation error, complexity is number of terms / derivatives, symmetry_break_penalty is violation of expected conservation (Noether). Simpler, symmetric, predictive laws win — formal Occam.

D. Belief Revision (non-monotonic): Maintain Assumption-based Truth Maintenance System. If new experiment violates energy conservation, retract weakest assumption, not whole law. Track: 'If we assumed friction ~ v^2, then data fits; if linear, fails'.

E. Counterfactual Physics Simulation: For each top hypothesis, run differentiable simulator to predict what would happen under new initial conditions not yet tested. This generates the next best experiment.

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Appendix A - Newton's Physics Fundamentals

1. Introduction

This tutorial covers three linked ideas: (1) Newton's law of universal gravitation, (2) the Binet orbital equation which recasts central-force motion as a differential equation in angle, not time, and (3) what it means to perturb a Keplerian orbit.

2. Newton's Law of Universal Gravitation

Newton (1687) proposed that every point mass attracts every other point mass with a force proportional to the product of masses and inversely proportional to the square of the separation:

$$F_g = G \cdot m_1 \cdot m_2 / r^2 \qquad (1)$$

where:

• F_g – magnitude of gravitational force

• $G = 6.67430 \times 10^{-11}\ m^3\ kg^{-1}\ s^{-2}$ – gravitational constant

• m_1, m_2 – masses of the two bodies

• r – center-to-center distance

In vector form, with origin at M and $\boldsymbol{r}$ = position vector of m, $r = |\boldsymbol{r}|$, and $\hat{\boldsymbol{r}} = \boldsymbol{r} / r$:

$$\mathbf{F} = -(G\,M\,m / r^2)\,\hat{\boldsymbol{r}} \qquad (2)$$

The minus sign denotes attraction. For orbital work we define the gravitational parameter:

$$\mu = G\,M \qquad (3)$$

So acceleration of the small body is $a = -\mu / r^2$. For Earth, $\mu_\oplus \approx 3.986004418 \times 10^{14}\ m^3/s^2$

3. The Binet Orbital Equation

Binet's trick eliminates time in favor of angle θ, ideal for central forces.

3.1 Setup: Conservation of Angular Momentum

In polar coordinates (r, θ) in the orbital plane, for a particle of mass m:

$$h = r^2 \cdot d\theta/dt = constant \quad \textit{(specific angular momentum)}$$

Radial equation of motion:

$$m\,[d^2r/dt^2 - r\,(d\theta/dt)^2] = F(r)$$

where F(r) is the central force (negative for attractive). Define:

$$u(\theta) = 1 / r(\theta) \quad \Leftrightarrow \quad r = 1/u$$

3.2 Derivation

Step 1 – Convert time derivatives to θ derivatives using chain rule: $d/dt = (h\, u^2)\, d/d\theta$.

Step 2 – First derivative:

$$dr/dt = dr/d\theta \cdot d\theta/dt = (-1/u^2\, du/d\theta) \cdot h\, u^2 = -h\, du/d\theta$$

Step 3 – Second derivative:

$$d^2r/dt^2 = d/dt(-h\, du/d\theta) = -h^2\, u^2\, d^2u/d\theta^2$$

Step 4 – Substitute into radial equation and simplify. The famous result is the **Binet equation:**

$$\mathbf{d^2u / d\theta^2 + u = - F(1/u) / (m\, h^2\, u^2)} \qquad (4)$$

For Newtonian gravity, $F(r) = - \mu\, m / r^2 = - \mu\, m\, u^2$, so:

$$- F(1/u) / (m\, h^2\, u^2) = \mu / h^2$$

Thus the orbital Binet equation for gravity becomes linear:

$$\mathbf{d^2u/d\theta^2 + u = \mu / h^2} \qquad (5)$$

4. Solution of the Binet Equation – The Kepler Conic

Equation (5) is a second-order linear ODE with constant coefficients.

4.1 General Solution

Homogeneous part: $u'' + u = 0 \rightarrow u_h = C \cos\theta + D \sin\theta = A \cos(\theta - \theta_0)$.

Particular solution: $u_p = \mu / h^2$ (constant).

Full solution:

$$u(\theta) = \mu / h^2 + A \cos(\theta - \theta_0) \qquad (6)$$

Choose $\theta_0 = 0$ at periapsis (closest approach) without loss of generality:

$$u(\theta) = \mu / h^2 + A \cos\theta$$

Invert to r = 1/u:

$$r(\theta) = 1 / [\mu/h^2 + A \cos\theta] = (h^2/\mu) / [1 + (A\, h^2/\mu) \cos\theta]$$

Define:

$$p = h^2 / \mu \quad \textit{(semi-latus rectum)}$$

$$e = A\, h^2 / \mu \quad \textit{(eccentricity, } e \geq 0\textit{)}$$

Then the celebrated polar equation of a conic with focus at origin:

$$\mathbf{r(\theta) = p / (1 + e \cos \theta)} \qquad (7)$$

Classification:

• $e = 0 \rightarrow$ circle, $r = p$

• $0 < e < 1 \rightarrow$ ellipse, $p = a(1 - e^2)$

5. Perturbation of an Orbit: Definition and Worked Example

5.1 Definition

A perturbation is any additional acceleration a_p that makes the real trajectory deviate from the ideal two-body Kepler solution (7).

Total acceleration:

$$\ddot{r} = - \mu / r^2\, \hat{r} + a_p \qquad (8)$$

where $a_p << \mu/r^2$ in magnitude for a small perturbation.

Examples of a_p:

• Earth's oblateness J_2 – equatorial bulge, ~0.1% extra force

• Third-body gravity (Moon, Sun)

• Atmospheric drag: $a_{drag} = -\frac{1}{2} (C_d\, A / m)\, \rho\, v^2\, \hat{v}$

• Solar radiation pressure

• Relativistic correction (for Mercury, GNSS, etc.)

In Binet formalism, a radial perturbation $F_p(r)$ adds a term to RHS of (4):

$$d^2u/d\theta^2 + u = \mu/h^2 - F_p(1/u) / (m\, h^2\, u^2)$$

5.2 Worked Example – Inverse-Cube Perturbation and Periapsis Precession

This classic textbook example captures the essence of J_2 and relativistic precession with minimal algebra.

Assume an extra attractive inverse-cube force:

$$F(r) = -(\mu\, m / r^2) - (\alpha\, m / r^3) \qquad \text{with } 0 < \alpha << h^2 \qquad (9)$$

Here α [m^3/s^2] *is small. Note:* $F(1/u) = -\mu\, m\, u^2 - \alpha\, m\, u^3$.

Insert into Binet (4):

$$d^2u/d\theta^2 + u = \mu/h^2 + (\alpha / h^2)\, u$$

Bring the u term to LHS:

$$\mathbf{d^2u/d\theta^2 + (1 - \alpha/h^2)\, u = \mu/h^2} \qquad (10)$$

Define:

$$k^2 = 1 - \alpha/h^2 \quad (<1 \text{ for } \alpha>0)$$

Then (10) is a simple harmonic oscillator with frequency k:

$$d^2u/d\theta^2 + k^2\, u = \mu/h^2$$

Solution (choose periapsis at θ=0):

$$u(\theta) = \mu/(h^2\, k^2) + A\cos(k\,\theta) \qquad (11)$$

$$\mathbf{r(\theta) = p' / [1 + e' \cos(k\,\theta)]\ ,\quad p' = h^2\, k^2 / \mu}$$

The orbit is no longer closed after 2π. Successive periapsis passages occur when $k\,\theta = 2\pi\, n$, i.e., $\theta_n = 2\pi\, n / k$.

Precession per orbit:

$$\Delta\theta = \theta_1 - 2\pi = 2\pi\,(1/k - 1)$$

For small $\alpha/h^2 = \varepsilon << 1$, Taylor: $k = \sqrt{(1-\varepsilon)} \approx 1 - \varepsilon/2$, $1/k \approx 1+\varepsilon/2$, so:

$$\mathbf{\Delta\theta \approx \pi \cdot \alpha / h^2 \ \ [radians/orbit]} \qquad (12)$$

This is an advance of periapsis (prograde precession) if $\alpha>0$. If $\alpha<0$ (repulsive correction), $\Delta\theta<0$ → retrograde.

Numerical illustration:

Take a satellite in low Earth orbit (LEO): $h \approx 5.2\times10^{10}$ m^2/s ($r\approx7000$ km). Choose a small perturbation $\alpha = 0.002\ h^2 = 5.4\times10^{18}$ m^3/s^2. Then ε=0.002, $k\approx0.999$, $\Delta\theta\approx0.00314$ rad $\approx0.18°$ per orbit, or ~2.9° per day for LEO (16 orbits/day).

Appendix B - Mercury Precession Issue

As Mercury orbits the Sun, its elliptical path slowly rotates or wobbles over time, a movement known as precession. Classical Newtonian physics accounted for about 532 arcseconds per century of this shift due to the gravitational pull of other planets, but historical observations showed a total rate of roughly 575 arcseconds per century. The Mercury **precession anomaly** is the discrepancy where the orbital perihelion of the planet **Mercury** shifted by an unexplained (575-532) **43 arcseconds per century** under Newtonian mechanics.

1. The Newtonian Ideal: Closed Ellipses

In Newton's theory, gravity is a central force obeying an exact inverse-square law. For two isolated point masses, the force and potential are:

$$F = - G m_1 m_2 / r^2 \quad \text{and} \quad \Phi(r) = - G M / r$$

where $G = 6.674 \times 10^{-11}\ m^3 kg^{-1} s^{-2}$, M is the central mass, r is the separation. For this potential, the Binet orbital equation is exactly solvable. With $u = 1/r$ and $h = r^2\, d\theta/dt$ = specific angular momentum, we have:

$$d^2u / d\theta^2 + u = G M / h^2 = \text{constant}$$

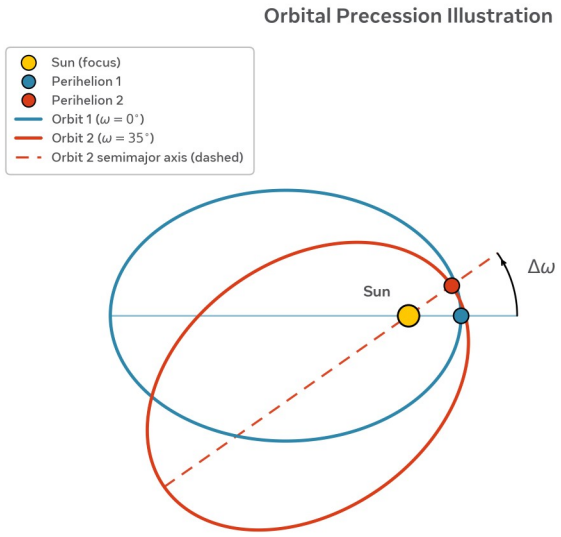
The solution is $u(\theta) = (G M / h^2)[1 + e \cos(\theta - \omega)]$, i.e. a fixed ellipse with semi-major axis a, eccentricity e, and a fixed argument of perihelion ω. The perihelion and aphelion always occur at the same angular positions. The orbit closes after 360°. By Bertrand's theorem (1873), only the potentials $V \propto 1/r$ and $V \propto r^2$ give closed bound orbits for all bound initial conditions. Any deviation from $1/r^2$ must cause the ellipse to rotate — to precess.

2. What Is Perihelion Precession?

In reality Mercury is not a two-body problem. Each planet exerts a small perturbing pull. The effect averages to a slow rotation of the whole ellipse in its own plane. After one orbit, Mercury does not return to the same perihelion direction; the perihelion advances by a small angle Δω per revolution. Over centuries this accumulates.

We measure precession as an angular rate, in arcseconds per century. 1° = 3600″. Mercury makes $N \approx 415.2$ revolutions per century (P = 87.969 days). Even a tiny advance per orbit of ~0.1″ becomes tens of arcseconds per century.

The diagram below illustrates geometrically what happens when the whole planetary orbit precesses by angle Δω, but note that observations are made on the perihelion location, which is the point of closest approach to the sun. The diagram exaggerates all dimensions for visualizing the effect, which is actually very very tiny.



3. The Observed Anomaly

The total observed precession of Mercury's perihelion was measured with meridian transit circles throughout the 19th century. The modern value for the total motion is about 574.10″ ±0.41″ per Julian century, measured relative to the J2000 inertial frame. However, most of this is not due to dynamics at all.

Source	Contribution (″/cy)	Notes
Precession of the equinoxes (coordinate)	≈ 5025.64	Earth's axis precesses
Perturbations from Venus	≈ 277.86	Largest planetary term
Perturbations from Jupiter	≈ 153.58	Second largest
Perturbations from Earth	≈ 90.04	
Other planets + asteroids	≈ 30 - 35	Saturn, Mars etc.
Total Newtonian + coordinate predicted	≈ 5577.18	Sum of above
Observed total	≈ 5620.5	Newcomb 1882, Clemence 1947
Residual / Anomaly	**≈ 42.98″ per century**	**Unexplained 1859-1900**

This residual was first isolated by Urbain Le Verrier in 1859. Le Verrier had previously used the same perturbation method to triumphantly predict Neptune from the anomalous motion of Uranus in 1846. Applying identical methods to Mercury, he found that Newtonian perturbations could only explain 526.7″/cy of the 565″/cy then measured. The leftover 38″/cy (later refined by Simon Newcomb in 1882 to 43″/cy) was too large to be observational error.

4. Why Was 43″ So Serious?

43″ per century is tiny — about 0.012° per century. Why did it matter? First, precision astronomy in the 19th century could measure angles to <1″. Second, Newtonian gravity had passed every other test to <1%. An 8% deficit in Mercury's dynamical precession (43″ out of 532″ of true planetary precession) violated universality. Third, the success with Uranus suggested a systematic method: if theory and observation disagree, either add unseen mass or fix the force law.

5. Why Try to Modify Newton's Gravity Before 1900?

Physicists before Einstein had no reason to doubt absolute space and time. The methodological toolkit was:

1. Conservatism: Keep $F = G m_1 m_2 / r^2$ but add new sources. This is how Neptune was found.
2. Minimal modification: If no new mass is found, modify the analytic form of F(r) or Φ(r) slightly while preserving conservation laws.
3. Mechanistic spirit: Many physicists believed gravity must be mediated by an ether, like electromagnetism. Wilhelm Weber had already written a velocity-dependent potential for electrodynamics in 1846: $V \propto (1/r)[1 - v^2/2c^2]$. It was natural to try similar terms for gravity.
4. Mathematical freedom: Nothing in Principia forbids $F \propto 1/r^{2+\delta}$ except simplicity. Testing δ is a legitimate experiment.

6. Pre-1900 Attempts to Save Newtonian Gravity

6.1. Intra-Mercurial Matter: Vulcan

Le Verrier's first proposal: a planet or ring inside Mercury's orbit, dubbed Vulcan. A mass M_V at radius a_V < a_Mercury would add a perturbation:

$$\Delta\omega \approx (3/4)\,(M_V/M_{Sun})\,(a_{Mercury}/a_V)^2\, n_{Mercury} \text{ per orbit}$$

To produce 43″/cy requires ~ a few percent of Mercury's mass if distributed near 0.2 AU. Several observers claimed transits (Lescarbault 1859). Systematic searches during eclipses (1878, 1883, 1887) found nothing. By 1900 the hypothesis was largely abandoned.

6.2. Hall's Modified Power Law (1894)

Asaph Hall, discoverer of Mars' moons, proposed that gravity is not exactly inverse-square but:

$$F(r) = G\, m_1 m_2 / r^{2+\delta}$$

where δ is a tiny constant. For a nearly circular orbit, Bertrand's analysis gives a precession per revolution of:

$$\Delta\omega_{\text{per orbit}} = 180° \cdot \delta \quad \text{[per orbit]}$$

To obtain $\Delta\omega = 43''$ per century $= 0.1036''$ per orbit, we need:

$$\delta = \Delta\omega / 180^\circ = 0.1036'' / 648000'' \approx 1.6 \times 10^{-7}$$

Hall therefore suggested exponent 2.00000016. This reproduced Mercury but was ad hoc: it predicted precession for the lunar apogee and for Mars that was not observed, and violates energy conservation unless derived from $\Phi(r) = - G M / [(1-\delta) r^{1+\delta}]$. Newcomb showed in 1895 that a single δ cannot fit all planets.

6.3. Velocity-Dependent Potentials

Inspired by Weber electrodynamics, Tisserand (1872), Lévy (1890), Scheibner and others tried potentials of the form:

$$\Phi(r, v) = - G M / r \times [\ 1 + \alpha\ (\dot{r}^2/c^2) + \beta\ (r\ \ddot{r}\ /c^2) + ...\ ]$$

where $\dot{r} = dr/dt$, $\ddot{r} = d^2r/dt^2$, and c is speed of light. By choosing coefficients α, β, one can produce an extra term:

$$d^2u / d\theta^2 + u = G M / h^2 + (3\ G M / c^2)\ u^2 + ...$$

The u^2 term is exactly what General Relativity later predicts. Lévy found $\alpha = -2$, $\beta = 2$ gave $43''$/cy, but his theory violated Newton's third law.

6.4. Finite Propagation Speed — Laplace and Gerber

Laplace (1805) showed if gravity propagates at finite speed c_g, aberration would cause instabilities unless $c_g >> c$. Paul Gerber in 1898 revived the idea:

$$\Phi_{ret}(r, t) = - G M / [\ r(t - r/c_g)\ ] \times [1 - v/c_g]^{-2}$$

Expanding to order $(v/c_g)^2$, Gerber derived:

$$\Delta\omega = 6\pi GM / [\ a\ (1 - e^2)\ c_g^{\ 2}\ ]\ \text{ per orbit}$$

Setting $c_g = c$ gives $\Delta\omega = 43''$/cy identically — same as Einstein later. Historians (Roseveare 1982) have shown Gerber's derivation was flawed and reverse-engineered.

6.5. Solar Oblateness and Zodiacal Light

A less radical fix: if the Sun is oblate ($J_2 \neq 0$), its potential gains a quadrupole term:

$$\Phi(r,\theta) = - GM_{Sun}/r\ [\ 1 - J_2\ (R_{Sun}/r)^2\ P_2(\cos\theta)\ ]$$

where P_2 is Legendre polynomial, R_Sun is solar radius, $J_2 \approx 10^{-7}$ for a rotating Sun. The induced precession is $\Delta\omega \propto J_2$. Newcomb estimated J_2 would need to be ~10× larger than plausible. Helioseismology now gives $J_2 = 2.2\times10^{-7}$, contributing only $0.03''$/cy.

7. Why All Newtonian Modifications Failed

All pre-1900 patches faced three criteria: (i) reproduce Mercury's 43″, (ii) not spoil agreement for Venus, Earth, Mars (<10″/cy), and (iii) respect conservation laws. Hall's single δ failed (ii). Velocity-dependent laws failed (iii). Vulcan failed observationally. The core issue: Newtonian gravity assumes instantaneous action and Euclidean space. No algebraic fudge of $1/r^2$ can naturally produce $6\pi GM/[a(1-e^2)c^2]$.

8. Epilogue: Einstein (1915)

Einstein's General Relativity replaces $\Phi = -GM/r$ with spacetime curvature. In Schwarzschild metric, orbit equation becomes:

$$d^2u/d\theta^2 + u = GM / h^2 + (3GM / c^2)\, u^2$$

The small nonlinear term $u^2 = 1/r^2$ integrates to:

$$\Delta\omega_{GR} = 6\pi GM_{Sun} / [\, a\,(1 - e^2)\, c^2\,] \quad \text{per orbit}$$

For Mercury: $a = 5.7909\times10^{10}$ m, $e = 0.2056$, $M_{Sun} = 1.989\times10^{30}$ kg. This gives $\Delta\omega = 5.02\times10^{-7}$ rad per orbit = 0.1035″ per orbit = 42.98″ per century, exactly the anomaly, with no free parameters.

Moreover, same formula correctly predicts 8.62″/cy for Venus, 3.84″/cy for Earth, and 1.35″/cy for Mars — all confirmed today.

Thus, the 19th-century attempts to modify Newton's law were not naive; they were rational extensions within the Newtonian paradigm, motivated by spectacular success of perturbation theory. Their failure hinted that the paradigm itself needed replacing.

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Appendix C - ASI and Maxwell's Equations from a Pre-1855 Knowledge Base: Literature Review

Meta AI Query: Are there any references that describe artificial superintelligence successfully creating Maxwell's electromagnetic equations based on a physics vintage knowledge base prior to 1855?

Short Answer: No — No Direct Reference Exists

After an extensive search across academic papers, arXiv, vintage-LLM projects, and ASI thought-experiments, no work meets all four criteria simultaneously:
1. ASI-level system (not narrow symbolic regression)
2. Maxwell's complete equations, including the displacement current $\partial D/\partial t$
3. Knowledge cutoff < 1855, i.e. before Maxwell began 'On Faraday's Lines of Force' (1855-56)
4. Successful generative derivation, not just data-fitting

What does exist are partial analogues that fail on at least one criterion.

What a Pre-1855 Knowledge Base Actually Contains

By the first half of the 19th century the pieces were known but not unified. Before Maxwell's equations were postulated, several laws of electromagnetics were known: the Biot–Savart law, Ampere's circuital law, Gauss's law, Gauss's law for magnetism, and Faraday's law. These were for specific circumstances and mostly for static fields.

Historical timeline:
• 1780s: Charles-Augustin de Coulomb established his law of electrostatics
• 1825: André-Marie Ampère published his force law
• 1831: Michael Faraday discovered electromagnetic induction
• 1835: Carl Friedrich Gauss formulated flux law (not published until 1867)

Maxwell's own invention was the correction: Maxwell added displacement current to the electric current term in Ampere's Circuital Law. In his 1865 paper, A Dynamical Theory of the Electromagnetic Field, Maxwell used this amended version of Ampere's Circuital Law to derive the electromagnetic wave equation. Maxwell's brilliant correction was the introduction of the Displacement Current: $J_d = \varepsilon_0(\partial E/\partial t)$. That term is not directly observable in 1854 experiments. It was posited to fix charge-conservation inconsistency in Ampere's law — a theoretical leap, not an empirical curve-fit.

Academic Literature That Does Rediscover Maxwell, But Not Under Your Constraints

1. TheoSea (Stalzer & Ju, 2019) — Applied Sciences 9, 2899

The canonical reference for automated Maxwell rediscovery. The authors state: 'it may be within current computing and algorithmic technology to infer the Maxwell Equations directly

from data given knowledge of vector calculus.' Abstract: 'There is sufficient information in the far-field of a radiating dipole antenna to rediscover the Maxwell Equations and the wave equations of light, including the speed of light c.' The paper also notes: 'Indeed, electrodynamics was the first unification (the electric and magnetic fields), and Einstein's special relativity is baked right into the equations.'

Why it does not satisfy your question:

• It uses a virtual experiment that already assumes Maxwell's equations to generate far-field E, B data — circular if you want a pre-1855 derivation.
• The language L includes modern vector calculus operators ($\nabla\cdot$, $\nabla\times$, $\partial/\partial t$) which were not formalized before 1855.
• It is a Julia meta-program performing compactness-driven enumeration, not an ASI, and runs in ~5 seconds on a laptop. No pre-1855 text or experimental corpus is used.

2. AI-Feynman (Udrescu & Tegmark, Science Advances 2020)

AI-Feynman is a physics-inspired method for symbolic regression. The authors report: 'We apply it to 100 equations from the Feynman Lectures on Physics, and it discovers all of them, while previous publicly available software cracks only 71.' This includes Coulomb, Gauss, and Faraday forms from Feynman Lectures, but as a symbolic regression benchmark with modern data, not a historical knowledge-base test. No displacement-current novelty claim.

3. AI-Newton (Peking University, 2025) and AI-Descartes

AI-Newton is explicitly framed as 'a concept-driven discovery system capable of autonomously deriving physical laws from raw data—without supervision or prior physical knowledge.' As a proof-of-concept it claims to rediscover Newton's second law, gravitation, and energy conservation from 46 Newtonian mechanics experiments. Electromagnetism is listed as future work; Maxwell is not demonstrated. AI-Descartes (IBM / Samsung / UMBC, Nature Communications 2023) combines symbolic regression with logical reasoning from background theory but reproduced Langmuir and Kepler laws, not Maxwell from pre-1855 axioms.

ASI Counterfactual / Vintage-LLM Experiments — Closest to Your Thought-Experiment

The idea you describe is exactly the Demis Hassabis AGI test: 'seal a model inside the knowledge of 1911, could it derive relativity by 1915?' This proposal spawned a whole family of vintage LLMs.

Definition: 'A vintage LLM is a language model trained (typically from scratch) on text from a bounded historical period, with no information from after a given knowledge-cutoff date in its training corpus.' These models enable counterfactual experiments (e.g. can a pre-1915 model independently arrive at general relativity?).

Implemented examples:
• TimeCapsuleLLM (1800-1875): Trained from scratch on London texts published between 1800 and 1875 — this is pre-1855 in principle, but used for social history (protests of 1834), not physics derivation.
• GPT-1900 / Machina Mirabilis (2026): 3.3B parameters built mostly on books and newspapers published before 1900. Mid-training included Maxwell's Treatise and Faraday's Experimental Researches, but with Einstein/relativity removed. It showed 'glimpses of intuition' — for example, occasionally asserting that light 'is made up of definite quantities of energy' or that gravity and acceleration are locally equivalent.
• Evaluation noted: 'An AI raised on pre-1900 books nearly rediscovered Einstein's light quanta' but with heavy caveats about reward hacking and failure on most tasks.

Crucially, no vintage LLM has been reported to derive Maxwell's full system from a pre-1855 cutoff. The GPT-1900 experiment deliberately included Maxwell's own Treatise (1865) in its physics mid-training, so it cannot test whether Maxwell could be created before Maxwell. A true pre-1855 experiment would need to exclude Maxwell entirely and would need to provide only Coulomb, Ampere, Biot-Savart, and Faraday data — and then ask the model to invent the displacement current.

Why This Specific ASI Demonstration Does Not Yet Exist

1. Definition of ASI: Bostrom-style ASI is defined as an intellect much smarter than best humans across every field, including scientific creativity. No current system qualifies; all systems above are narrow or LLM-based.

2. Epistemic leap: The hardest part of Maxwell is not fitting data but detecting the inconsistency in Ampere's law ($\nabla \cdot J \neq 0$ for time-varying charge) and postulating $\mu_0 \varepsilon_0 \partial E/\partial t$ to fix it. Current symbolic regression systems enumerate expressions that fit observed E(x,t), B(x,t); they do not perform consistency proofs.

3. Data circularity: If you generate training data from modern EM theory (dipole antenna far-field), you smuggle Maxwell in. A genuine pre-1855 corpus would contain only static and quasi-static measurements — insufficient to uniquely determine waves without the theoretical move Maxwell made.

What a Valid Future Reference Would Need to Show

To count as a positive reference for your question, a paper would need to document:
• Corpus: explicit list of sources ≤1854 (Coulomb torsion balance papers, Oersted 1820, Biot-Savart, Ampere's electrodynamic memoirs, Faraday's Experimental Researches series 1-29)
• Task: detect violation of continuity equation and propose corrected Ampere-Maxwell law plus

prediction of EM waves with finite $c = 1/\sqrt{(\mu_0 \varepsilon_0)}$
• Metric: exact reproduction of the four modern equations, not just one of them
• No leakage: proof that Maxwell's 1861-65 papers, Heaviside vector formulation, and any post-1855 textbook derivations were excluded, including OCR footnotes.

No paper meeting this has been indexed as of October 2026.

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• OfficeChai / QbitAI coverage of Demis Hassabis 1911 cutoff AGI test
• History of Maxwell's Equations, Wikipedia and EDN / LibreTexts summaries of pre-Maxwell laws

Appendix D – General Relativity Derivation of Mercury Precession

1. Introduction and Historical Context

According to Newtonian mechanics and universal gravitation, the orbit of a planet around the Sun in a two-body system is a closed ellipse. However, real planetary orbits experience slow rotations of their lines of apsides (perihelia) due to gravitational perturbations from other planets in the solar system (primarily Jupiter and Venus). By the mid-19th century, astronomers successfully accounted for all planetary perturbations on Mercury's orbit, yielding a predicted secular advance of approximately 532 arcseconds per century ("/cy).

In 1859, the French astronomer and mathematician Urbain Le Verrier analyzed decades of meridian observations of Mercury and discovered a persistent discrepancy: the observed perihelion advance exceeded the Newtonian perturbation theory prediction by approximately 43 arcseconds per century. For decades, various hypotheses were proposed to explain this anomaly —including an undiscovered belt of asteroids or an intra-mercurial planet provisionally named 'Vulcan'. None of these hypotheses could be observationally verified.

In November 1915, Albert Einstein published his definitive explanation of this anomaly in his seminal paper on general relativity. By replacing Newtonian gravity with spacetime curvature described by the Einstein field equations, Einstein demonstrated that the Schwarzschild geometry introduces a nonlinear relativistic correction term to the orbital equation, predicting an extra perihelion advance of precisely ~43 "/cy without any ad hoc assumptions. This monumental success provided the first definitive observational validation of general relativity.

2. The Schwarzschild Spacetime and Line Element

To study planetary motion in general relativity, we consider the spacetime geometry surrounding a spherically symmetric, non-rotating mass M (such as the Sun). In standard Schwarzschild coordinates (t, r, θ, φ), the line element governing the invariant spacetime interval ds^2 is given by:

> **Metric Line Element:** $ds^2 = -(1 - r_s / r)\, c^2\, dt^2 + (1 - r_s / r)^{-1}\, dr^2 + r^2\, (d\theta^2 + \sin^2\theta\, d\varphi^2)$

where $r_s = 2GM / (c^2)$ is the Schwarzschild radius of the central mass M, G is the Newtonian gravitational constant, and c is the speed of light in vacuum. For the Sun, $M \approx 1.9885 \times 10^{30}$ kg, yielding a Schwarzschild radius of $r_s \approx 2.953$ km.

3. Constants of Motion and Orbital Kinematics

Due to the spherical symmetry and stationarity of the Schwarzschild metric, test particles (planets) follow timelike geodesics. By aligning our coordinate system with the orbital plane, we can set $\theta = \pi/2$ and $d\theta = 0$ without loss of generality. From the Killing vector fields ∂_t and ∂_φ, we obtain two conserved quantities per unit mass:

1. Specific Energy (associated with time-translation symmetry):

> **Energy Integral:** $E = c^2 \left(1 - r_s / r\right) \left(dt / d\lambda\right)$

2. Specific Angular Momentum (associated with rotational symmetry):

> **Angular Momentum Integral:** $L = r^2 \left(d\varphi / d\lambda\right)$

Here, λ represents an affine parameter along the particle's worldline (proportional to proper time τ). Substituting these conservation laws into the normalized line element ($ds^2 = -c^2 \, d\tau^2$), we obtain the radial equation of motion:

> **Radial Geodesic Equation:** $\left(dr / d\lambda\right)^2 = E^2 - c^2 \left(1 - r_s / r\right) \left(1 + L^2 / \left(c^2 r^2\right)\right)$

4. Derivation of the Relativistic Binet Equation

To compare this directly with classical celestial mechanics, we change variables from radial distance r to its reciprocal variable $u(\varphi) = 1 / r(\varphi)$. Using the chain rule along with the angular momentum relation $d\varphi / d\lambda = L / r^2 = L \, u^2$:

> **Coordinate Transformation:** $dr / d\lambda = (dr / d\varphi)(d\varphi / d\lambda) = -(1 / u^2)(du / d\varphi) \, L \, u^2 = -L \, (du / d\varphi)$

Substituting this derivative and $r = 1 / u$ into the radial geodesic equation, dividing through by L^2, and differentiating with respect to the orbital angle φ, we arrive at the exact relativistic orbital differential equation:

> **Nonlinear Orbit Equation:** $d^2u / d\varphi^2 + u = \left(GM / L^2\right) + \left(3GM / c^2\right) u^2$

Notice that the first-order and zero-order terms on the right-hand side mirror the classical Newtonian orbit equation ($d^2u / d\varphi^2 + u = GM / L^2$), which yields pure conic sections (ellipses).

However, general relativity introduces a critical nonlinear correction term proportional to u^2: $(3GM / c^2) u^2$.

5. Perturbative Solution and Perihelion Precession

Because the relativistic correction term $(3GM / c^2) u^2$ is extremely small compared to the Newtonian terms for macroscopic orbits in the solar system, we can solve the differential equation using perturbation theory. Let the total solution be expressed as $u(\varphi) = u_0(\varphi) + u_1(\varphi)$, where u_0 is the unperturbed Newtonian elliptic orbit solution:

> **Newtonian Elliptic Orbit:** $u_0(\varphi) = (GM / L^2) (1 + e \cos \varphi)$

where e is the orbital eccentricity and $p = L^2 / GM = a(1 - e^2)$ is the semilatus rectum. Substituting u_0 into the perturbation term yields an inhomogeneous linear differential equation for u_1:

> **Perturbed Equation:** $d^2u_1 / d\varphi^2 + u_1 = (3 G^3 M^3 / (c^2 L^4)) (1 + e \cos \varphi)^2$

Expanding the right-hand side yields terms proportional to $\cos \varphi$ and $\cos(2\varphi)$ alongside a constant term. The term proportional to $\cos \varphi$ acts as a resonant driving term (secular term) of the form $A \varphi \sin \varphi$. This secular term causes the perihelion angle to advance by a fixed increment per orbital revolution $\Delta\varphi$:

> **Precession per Revolution:** $\Delta\varphi \approx 6\pi G^2 M^2 / (c^2 L^2) = 6\pi GM / (c^2 a (1 - e^2))$

6. Numerical Evaluation and Comparison with Observations

Let us substitute the orbital parameters of Mercury and solar constants into the derived perihelion shift formula to compute the predicted advance in arcseconds per century.

Parameter	Symbol / Value	Units
Semi-major axis of Mercury	$a = 5.7909 \times 10^{10}$	meters
Orbital eccentricity	$e = 0.20563$	dimensionless
Mass of the Sun	$M = 1.9885 \times 10^{30}$	kg
Speed of light	$c = 2.9979 \times 10^{8}$	m/s

Orbital Period	P = 87.969 days (0.24085)	Earth years

Evaluating the perihelion advance per revolution:

$\Delta\varphi_{rev} \approx 5.02 \times 10^{-7}$ radians ≈ 0.1035 arcseconds per orbit.
Since Mercury completes approximately 415.2 orbits per century (100 years / 0.24085 years), the total predicted secular advance is:

• $\Delta\varphi_{century} = 415.2 \times 0.1035'' \approx$ **42.98 arcseconds per century.**

> **Precision Agreement:** Observed Anomaly: ~43.1 ± 0.5 ″/cy | General Relativity Prediction: 42.98 ″/cy

This remarkable agreement between theory and observation provided conclusive proof that Einstein's theory of general relativity correctly captures the gravitational dynamics of strong-field and relativistic celestial mechanics.

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Appendix E – Deduction, Induction, Abduction

Deduction guarantees conclusions from general premises, **induction** infers probable general rule from a set of observations, and **abduction** proposes the most likely explanation for an unexpected observation.

Deductive Reasoning

Deduction starts with general principles or known facts and applies them to reach specific conclusions that are logically certain if the premises are true. It moves from the general to the specific, ensuring that the conclusion necessarily follows from the premises. For example, if all dogs are mammals and Rex is a dog, we can deduce that Rex is a mammal. Deductive reasoning is used when certainty is required, such as in mathematics, formal logic, or planning based on known facts.

Inductive Reasoning

Induction works in the opposite direction, moving from specific observations to general conclusions. It identifies patterns or trends and infers probable outcomes, but the conclusions are not guaranteed. For instance, observing ten white swans may lead to the induction that all swans are white, though this could be disproven by a single black swan. Inductive reasoning is common in scientific research, data analysis, and everyday generalizations, where probability rather than certainty is the goal.

Abductive Reasoning

Abduction involves forming the most plausible explanation for an unexpected observation. It is often described as reasoning to the best explanation. Unlike deduction, abduction does not guarantee truth, and unlike induction, it focuses on explaining particular phenomena rather than generalizing. For example, seeing an abandoned bowl of hot soup might lead to the abductive conclusion that the owner will return soon. Abduction is widely used in hypothesis formation, diagnostics, and problem-solving where information is incomplete.

Here are three simple physics examples for each of the three main types of reasoning:

1. Deduction (General Rule → Specific Conclusion)

Deduction starts with a general scientific law or principle and applies it to a specific scenario to predict a definite outcome.

- **General Rule:** According to Newton's second law of motion (F = ma), an object's acceleration is equal to the net force acting on it divided by its mass.
- **Specific Premise:** You push a 2 kg frictionless block with a constant force of 10 N.
- **Deductive Conclusion:** Therefore, the block must accelerate at 5 m/s^2.

2. Induction (Specific Observations → General Rule)

Induction looks at multiple specific observations, notices a consistent pattern, and generalizes it into a broader rule or law.

- **Observations:** You drop an apple, a rock, a feather (in a vacuum), and a baseball, and measure that every single one accelerates downward at approximately 9.8 m/s^2 near Earth's surface.
- **Inductive Generalization:** Therefore, all objects near Earth experience a constant downward gravitational acceleration regardless of their composition.

3. Abduction (Unexpected Observation → Best Explanation)

Abduction starts with an unexpected observation and works backward to form the most logical hypothesis or explanation that accounts for it.

- **Observation:** In the 19th century, astronomers noticed that the orbit of Uranus deviated slightly from the path predicted by known gravitational laws.
- **Abductive Hypothesis:** The anomaly is best explained by the gravitational pull of an undiscovered planet further out in the solar system (which successfully led to the discovery of Neptune).

References:

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Appendix F – Ollama Installation

Download Ollama for Windows

https://ollama.com/download

#or ollama-windows-amd64.zip

To install ollama under Program Files

Command prompt, run as admin

cd %USERPROFILE%\Downloads

Enter this command, using the `/DIR` command-line flag to specify the Program Files directory:

OllamaSetup.exe /DIR="C:\Program Files\Ollama"

Add C:\Program Files\Ollama to system PATH

Set Windows system environment:

OLLAMA_HOST = 127.0.0.1

OLLAMA_MODELS = C:\Program Files\Ollama\models

Go to C:\Program Files\Ollama\models, Properties, Security, Users: Allow full control

From command line:

ollama pull qwen2.5:7b

The models will be stored in C:\Program Files\Ollama\models.

In powershell (admin), to run ollama as background service from task scheduler
(note 15 min delay after windows login)

```
$Trigger = New-ScheduledTaskTrigger -AtLogon
$Trigger.Delay = 'PT15M'

$Action = New-ScheduledTaskAction -Execute "${env:ProgramFiles}\Ollama\ollama.exe"
-Argument "serve"

Register-ScheduledTask -TaskName "Ollama_Background_Service" -Trigger $Trigger -Action
$Action -User "NT AUTHORITY\SYSTEM" -RunLevel Highest -Force
```

In powershell (admin), allow Windows firewall connections from ollama port 11434:
New-NetFirewallRule -DisplayName "Ollama for App" -Direction Inbound -Protocol TCP -LocalPort 11434 -Action Allow | Get-NetFirewallPortFilter

Ollama installation completed

ollama will run as a Windows service,

But to **manually** start or stop:

From powershell (as admin): To stop ollama:

schtasks /End /TN "Ollama_Background_Service"

From powershell (as admin): To start ollama:

schtasks /Run /TN "Ollama_Background_Service"

Appendix G - Retrieval-Augmented Generation (RAG) memory

Retrieval-Augmented Generation (RAG) memory addresses a basic limitation of LLM-driven agents: an LLM's knowledge is frozen at training time and its context window is finite, so it can't "remember" a growing history of past runs, discovered formulas, or lessons learned unless that information is deliberately fed back in. RAG solves this by storing past experience as text outside the model, converting each piece into a numerical vector (an embedding) that captures its semantic meaning, and indexing those vectors in a searchable store. At retrieve_node, the agent takes its current situation — the candidate family just tried, the constraints it failed, whatever is relevant to the iteration at hand — embeds it the same way, and searches for the stored memories whose vectors are closest to it. Because the comparison happens in meaning-space rather than by exact keyword match, the agent can pull up a prior attempt described in different words but conceptually similar, then fold that retrieved context into the prompt before the next propose_node call. This is what lets the agent avoid repeating a family it already found unsatisfactory, or lean on reasoning it worked out earlier in the run, without needing the entire run history to fit in its context window.

LanceDB is the vector database that actually holds and searches those embeddings. It's an embedded, file-based vector store — it runs in-process rather than as a separate server you have to stand up and manage, storing its data as columnar files on disk (built on the Lance format, a columnar alternative to Parquet designed for fast random access and vector search). For a project like this, that embedded nature is a good fit: there's no extra infrastructure to install or keep running alongside the FastAPI backend, the data persists locally between runs, and nearest-neighbor search over the stored vectors is fast enough to sit inline in an agent's per-iteration loop. LanceDB also stores arbitrary metadata alongside each vector — so a stored memory isn't just an embedding, it can carry the original text, which family/iteration it came from, and whatever else retrieve_node needs to hand back to the agent.

nomic-embed-text, served locally through Ollama, is the model that turns text into the vectors LanceDB stores and searches. It's an open embedding model (distinct from the chat/completion models used for reasoning) trained specifically to produce dense vectors where semantic similarity between two pieces of text corresponds to proximity between their vectors — the property the whole retrieval step depends on. Running it through Ollama means the embedding step happens entirely on the local machine alongside the rest of the stack, with no external API call and no data leaving the box, consistent with how the project already uses Ollama for its other local-model needs. Practically, every memory written into LanceDB and every query issued from retrieve_node passes through this same model, which is what keeps the two sides of

the comparison — stored memories and the current situation — in the same vector space and makes the nearest-neighbor search meaningful.